

MVE441 Statistical learning for big data

Project 2 - Feature selection and classification of Cat/Dog images

Hadi Chouman¹ Wilhelm Friberg¹ Zeeshan Khan¹
chouman@chalmers.se wilfrib@chalmers.se zeeshank@chalmers.se

¹Chalmers University of Technology

10 May 2026

Table of Contents

1 Classifiers

2 Feature selection

3 Flipped images

Table of Contents

1 Classifiers

2 Feature selection

3 Flipped images

The classifiers selected for modeling using the reduced feature space were:

- kNN (Neighborhood = 5, metric = euclidean, weights = distance)
- Logistic Regression ($C=0.1$)
- Random Forest (n estimators = 100)

The classifiers were selected arbitrarily. All methods used the same classifiers to make the feature selection methods more comparable.

Table of Contents

1 Classifiers

2 Feature selection

3 Flipped images

Feature selection - pre-processing

The original data size: (10000, 4096). To make the feature selection and classification more manageable, 2000 data points were used in the pipeline (1000 cat and 1000 dog images)

Three separate approaches and techniques for feature selection were considered:

- Combinatorial (Filtering + Lasso/Ridge)
- T-test + Forward selection
- T-test + Top-K Feature (Pixel) Selection

Feature selection - Combinatorial (Filtering + Lasso/Ridge)

The approach does CV for both filtering and Lasso/Ridge for each of the classifiers:

- Filtering methods: Mutual Information and F-score (ANOVA) (Top K selection)
- CV folds = 3
- Parameter grid search on filtering and Lasso/Ridge combination (Standardized data)
- Pipeline run 5 times to show accuracy spread and parameters

Feature selection - Combinatorial (Filtering + Lasso/Ridge)

Results from the 5 runs of the pipeline and the most common pixel selection

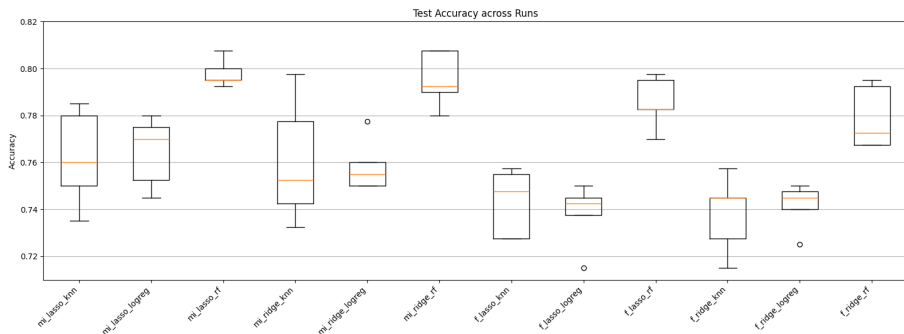


Figure: Boxplots of accuracies from all combinations of filtering and Lasso/Ridge feature selection and all classifiers

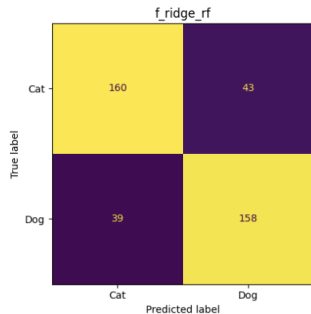
Feature selection - Combinatorial (Filtering + Lasso/Ridge)

Best performing parameters (majority vote across all runs):

- MI + F-score pixels: 250 px
- Lasso hyperparameter: $\alpha = 0.001 / \alpha = 0.01$
- Ridge hyperparameter: $\alpha = 0.0001 /$, threshold = 0.001

```
mi_lasso_knn: 43 pixels selected
mi_ridge_knn: 222 pixels selected
f_lasso_knn: 146 pixels selected
f_ridge_knn: 230 pixels selected
mi_lasso_logreg: 231 pixels selected
mi_ridge_logreg: 247 pixels selected
f_lasso_logreg: 239 pixels selected
f_ridge_logreg: 249 pixels selected
mi_lasso_rf: 150 pixels selected
mi_ridge_rf: 248 pixels selected
f_lasso_rf: 146 pixels selected
f_ridge_rf: 249 pixels selected
```

(a) 5th run pixels selected



(b) CM of F-score+Ridge using RandomForest classifier

Feature selection - Combinatorial (Filtering + Lasso/Ridge)

Figures shows the features selected first after filtering and filtering+Lasso/Ridge (Most common parameters used)

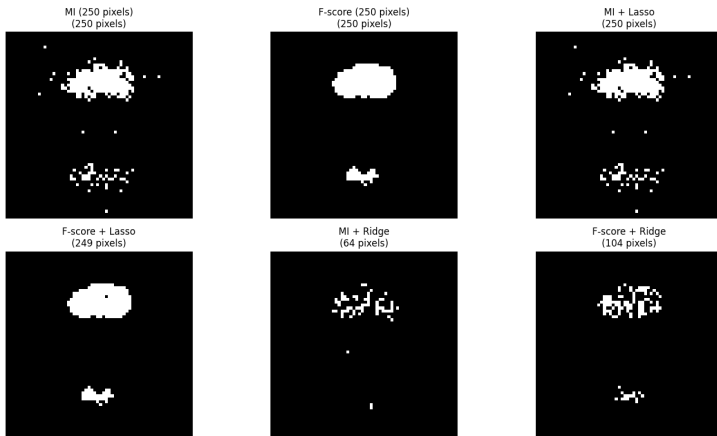
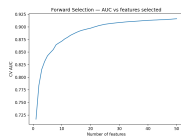


Figure: Caption

Feature selection - T-test + Forward selection

This approach has 3 steps:

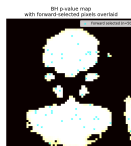
- Welch's T-test Benjamini-Hochberg ($\approx 1400/4096$)
- Random forest pruning 100 trees ($\approx 700/4096$)
- Sequential forward selection ($= 50/4096$)



(a) Forward selection performance curve (logistic classifier $cv=3$).



(b) Pixels selected after the T-test and after final forward selection: $4096 \rightarrow 1400 \rightarrow 700 \rightarrow 50$.



```
Random Forest AUC: 0.893
[[237  63]
 [ 50 250]]

KNN AUC: 0.828
[[263  37]
 [121 179]]

Logistic Regression AUC: 0.828
[[227  73]
 [ 76 224]]
```

(c) Accuracy results of the classifiers.

Figure: Results of the pipeline in one run.

Feature selection - T-test + Forward selection multiple runs

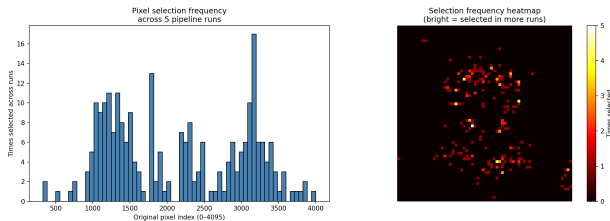


Figure: Which pixels were chosen across 5 runs.

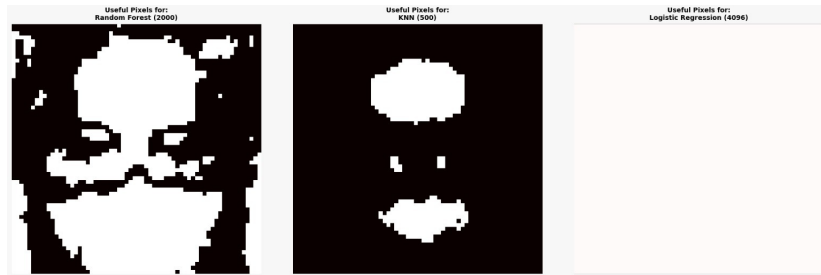
Run	T-test	RF	SFS
1	1366/4096 pixels	683/1366 pixels	50/683 pixels
2	1365/4096 pixels	683/1365 pixels	50/683 pixels
3	1443/4096 pixels	722/1443 pixels	50/722 pixels
4	1449/4096 pixels	725/1449 pixels	50/725 pixels
5	1353/4096 pixels	677/1353 pixels	50/677 pixels

Table: Number of pixels selected in each run.

Feature selection - T-test + Top-K Feature (Pixel) Selection

- T-test used to measure how different each pixel is between cat and dog images by comparing the average pixel values across the two classes; pixels with higher absolute t-scores are considered more discriminative.
- A 5-fold cross-validation was performed on the training set
- For each model, different numbers of top-ranked pixels (10, 50, 100, 500, 1000, 2000, 4096) were tested, and the subset with the highest cross-validation accuracy was selected.
- The selected pixel subset for each model was then used to train on the full training set and evaluated on the held-out test set. This entire procedure was repeated over 5 different random train-test splits to ensure robustness of results.
- Finally, a majority vote across the 5 runs was used to determine the final optimal number of pixels per model

Feature selection - T-test + Top-K Feature (Pixel) Selection



Pixel maps showing the most informative pixels selected for each model, based on the best cross-validation accuracy, and used for final evaluation on the test set. Results shown here come from one specific iteration.

Feature selection - T-test + Top-K Feature (Pixel)

Selection - CV Accuracy

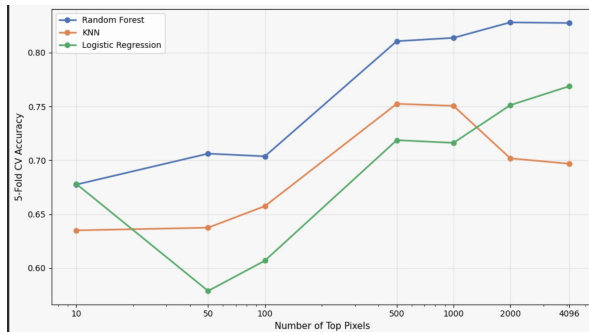


Figure: Cross-validation accuracy as a function of the number of top pixels selected by the t -test, for each model (Random Forest, KNN, Logistic Regression). Results shown here come from one specific iteration.

Feature selection - T-test + Top-K Feature (Pixel)

Selection - Confusion Matrix

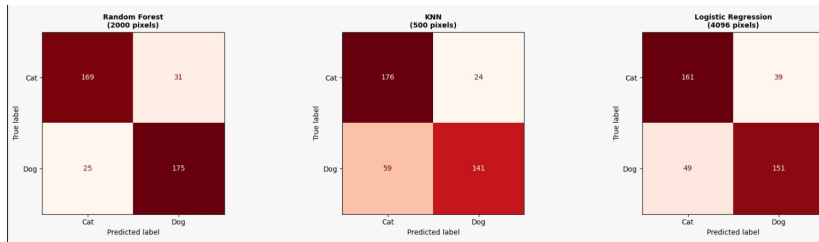


Figure: Confusion matrices showing the final test-set performance of each model trained using the top pixels that achieved the best cross-validation accuracy. Results shown here come from one specific iteration.

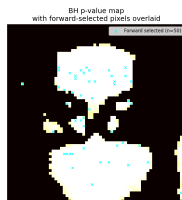
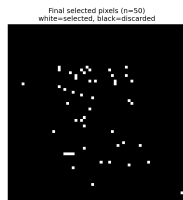
Table of Contents

1 Classifiers

2 Feature selection

3 Flipped images

Flipped images - T-test + Forward selection



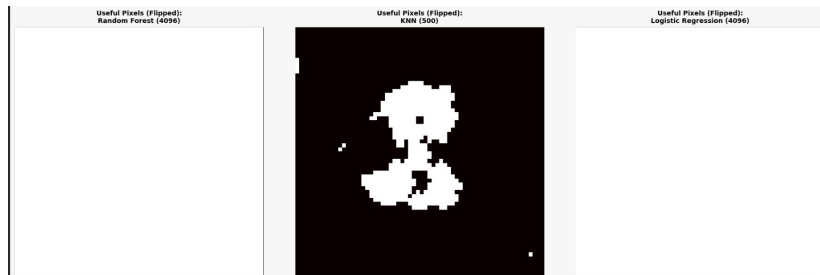
(a) Pixels selected after the T-test and after final forward selection (flipped):
4096 $\rightarrow$ 1327 $\rightarrow$ 664 $\rightarrow$ 50.

```
Random Forest  AUC: 0.865  
[[235  65]  
 [ 65 235]]  
  
KNN  AUC: 0.844  
[[191 109]  
 [ 37 263]]  
  
Logistic Regression  AUC: 0.839  
[[221  79]  
 [ 73 227]]
```

(b) Accuracy results for the classifiers on flipped images.

Figure: Results of the pipeline on flipped images in one run.

Flipped images - T-test + Top-K Feature (Pixel) Selection



Pixel maps showing the most informative pixels selected for each model, based on the best cross-validation accuracy, and used for final evaluation on the test set. Results shown here come from one specific iteration.

Flipped images - T-test + Top-K Feature (Pixel) Selection - CV Accuracy

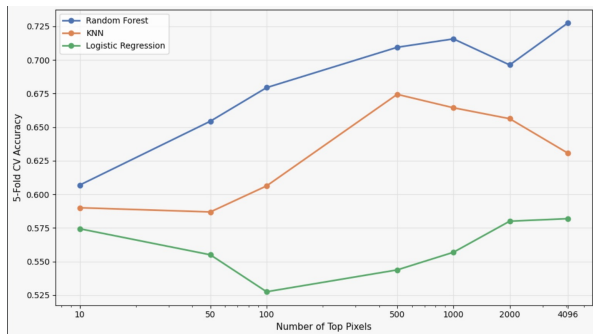


Figure: Cross-validation accuracy as a function of the number of top pixels selected by the t -test, for each model (Random Forest, KNN, Logistic Regression). Results shown here come from one specific iteration.

Flipped images - T-test + Top-K Feature (Pixel) Selection

- Confusion Matrix

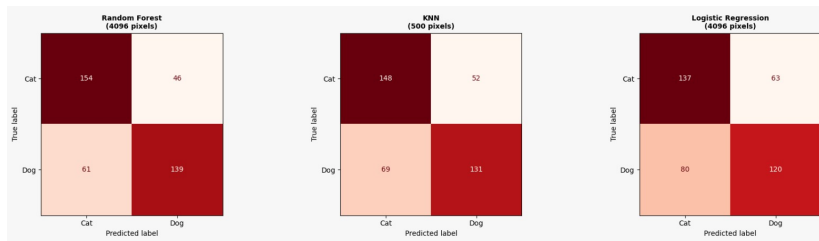


Figure: Confusion matrices showing the final test-set performance of each model trained using the top pixels that achieved the best cross-validation accuracy. Results shown here come from one specific iteration.